

Cheng-Yao (Eric) Wang

Research Scientist, Global Technology Applied Research, JPMorganChase

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Summary

I am an **XR and Human-AI Interaction Research Scientist** at **JPMorganChase**, where I design, prototype, and evaluate AI-powered systems grounded in real user workflows and measurable organizational outcomes. My work operates at the intersection of **HCI research, LLM/agent architecture, and end-to-end system development**, bridging research insight with working proof-of-concept solutions and production-oriented systems. I build AI-integrated systems that combine **LLMs, multimodal models, and agent frameworks** (e.g., LangGraph, AutoGen) with cross-platform interfaces (web, mobile, XR). My projects span problem framing, prompt and agent architecture design, system implementation, and evaluation, with a focus on creating AI systems that are **interpretable, actionable, and grounded in real workflows**.

My recent work explores **agent-assisted workflows, generative UI, and cognitive observability** for AI systems, focusing on how multi-agent pipelines can support complex tasks while remaining transparent and verifiable. My work has been published in **ACM CHI, CSCW, and IEEE ISMAR**. I received a **PhD in Information Science from Cornell University**, with prior training in Computer Science.

Research Interests

- **Agentic AI systems and workflows**, including multi-agent coordination, tool use, and LLM-driven task execution
- **Generative UI and AI-powered user experiences** that translate model capabilities into interactive systems
- **Human-AI collaboration in professional settings**, including workflow augmentation and decision support
- **End-to-end AI system prototyping and evaluation**, including prompt architecture, agent pipelines, and empirical validation

Skills

- **AI / LLM & Agent Systems:** prompt architecture, context engineering, LangGraph, AutoGen, multi-agent workflows, tool use, function calling
- **Generative UI & Interaction Systems:** LLM-driven UI generation, interactive prototypes, cross-platform AI interfaces (web, XR)
- **HCI Research & System Design:** translating user insights into system requirements, grounding AI behavior in workflows, empirical evaluation
- **Prototyping & Full-Stack Development:** end-to-end AI systems, API integration, rapid prototyping, deployment-ready architectures
- **Evaluation & Observability:** workflow metrics, task performance, cognitive load estimation, interaction analysis
- **Programming:** Python, JavaScript/TypeScript, Swift, C#, Kotlin, C++

Experience

Research Scientist (XR & Applied AI), JPMorganChase – New York, NY, USA Jan 2023 – present

- Research Scientist in Global Technology Applied Research, leading applied research at the intersection of **Human-AI interaction, agent systems, and immersive interfaces**.
- Designed and built **agent-assisted AI systems** integrating LLMs/VLMs with user interfaces to support complex workflows such as analysis, coordination, and decision-making.
- Developed **multi-agent pipelines and orchestration frameworks** (e.g., AutoGen, LangGraph) for task execution, tool use, and structured reasoning across real-world scenarios.
- Built systems such as **CognitiveSuite**, enabling cognitive observability of AI agent interactions through analysis of user behavior, task complexity, and interaction patterns.
- Led **end-to-end projects**, including stakeholder problem framing, system architecture, prototype implementation, and mixed-methods evaluation in real workflows.

Research Intern, Microsoft Research – Redmond, WA, USA June 2022 – Aug 2022

- Worked with Dr. Mar Gonzalez-Franco and Dr. Andy Wilson (EPIC team).
- Conducted research on VR/MR avatars and collaborative interaction techniques.

Research Intern, Meta Reality Labs – Toronto, Canada Sept 2021 – Feb 2022

- Collaborated with Dr. Mark Parent and Marcello Giordano.
- Developed Remote AR Transformer techniques for interaction across dissimilar physical spaces.

Research Intern, Microsoft Research – Redmond, WA, USA June 2021 – Aug 2021

- Worked with Dr. Mar Gonzalez-Franco, Dr. Daniel McDuff, and Dr. Eyal Ofek (EPIC team).
- Implemented CityLifeSim, a high-fidelity pedestrian and vehicle simulation system.

Research Intern, Autodesk Research – Toronto, Canada Jan 2021 – Mar 2021

- Worked with Dr. Fraser Anderson and Dr. Qian Zhou (HCI & VIS group).
- Developed and evaluated VideoPoseVR for avatar animation from online videos.

Education

Cornell University, PhD in Information Science – Ithaca, NY, USA Sept 2016 – Aug 2023

- Thesis: Sharing Transformed Experiences Across Time and Space through VR/MR and Transformation Design
- Advisor: Prof. Andrea Stevenson Won

National Taiwan University, MS in Computer Science – Taipei, Taiwan Feb 2012 – Aug 2014

- Advisor: Prof. Mike Y. Chen

National Taiwan University, BS in Computer Science – Taipei, Taiwan Sept 2008 – Jan 2012

Selected Projects

CognitiveSuite: Cognitive Observability for AI Agent Interactions

Jan 2026

Framework and prototype that augments LLM agent observability with cognitive and behavioral signals, enabling analysis of user experience, task difficulty, and interaction dynamics at scale.

- Designed a cognitive observability layer that enriches Phoenix traces with task complexity, reprompt patterns, and sentiment signals.
- Built pipeline to extract behavioral insights from real interaction data without requiring additional instrumentation from application teams.
- Integrated multi-signal analysis (prompt structure, sentiment, and typing-derived metrics such as rollover and error rate) to estimate user cognitive load.
- Developed interactive dashboard visualizing conversation flow, task diversion, and user state transitions to support adaptive agent design.

S-TIER: Situated-Traceable Insights in XR for Hybrid Crisis Management

Jan 2025

Framework and prototype that converts crisis-team conversation and multimodal context into traceable, situated XR insights for hybrid and asynchronous response work.

- Built an end-to-end pipeline from capture to structured insight generation and XR presentation.
- Integrated LLM/VLM reasoning with multimodal capture and spatial visualization.
- Designed for traceability: insights link back to source conversations and screen context.
- Evaluated in simulated crisis scenarios with distributed response teams.

From Discussion to Action: Delegated Work in XR Meetings

Jan 2026

XR meeting system enabling participants to delegate real tasks to agents during live discussions while preserving shared attention and collaboration flow.

- Designed agent-augmented XR interaction model exposing task state, progress, and clarification requests.
- Built automation agents using AutoGen plus Playwright and a parameterized DSL for reusable workflows.
- Developed Jupyter agent for live analysis and visualization driven by meeting context.

Entanglytics Playground: Agentic Workflow for Immersive Visualization

Jan 2026

AI-assisted system that converts natural language into executable 3D visualization code and renders results live across desktop and XR devices.

- Designed two-pass LLM pipeline with planning and constrained generation to improve reliability.
- Implemented runtime validation and type checking to ensure generated code matches APIs.
- Built cross-platform system delivering live visualizations to browsers and XR headsets.

Publications

SocialMiXR: Facilitating Hybrid Social Interactions at Conferences

May 2025

Cheng-Yao Wang*, Fannie Liu*, William Moriarty, Feiyu Lu, Usman Mir, David Saffo, Mengyu Chen, Blair MacIntyre

dl.acm.org/doi/10.1145/3711069 (PACM HCI (CSCW 2025))

Growing Together at Work: Cultivating a Mentorship Garden

May 2025

Erica Principe Cruz, Cheng-Yao Wang, William Moriarty, Blair MacIntyre, Fannie Liu

dl.acm.org/doi/10.1145/3710988 (PACM HCI (CSCW 2025))

Adaptive Content Placement in Mixed Reality Through Empirical User Behavioral Patterns Feiyu Lu, Mengyu Chen, Hsiang Hsu, Pranav Deshpande, Cheng-Yao Wang, Blair MacIntyre ieeexplore.ieee.org/document/10765314 (ISMAR Adjunct 2024)	Oct 2024
Adaptive 3D UI Placement in Mixed Reality Using Deep Reinforcement Learning Feiyu Lu, Mengyu Chen, Hsiang Hsu, Pranav Deshpande, Cheng-Yao Wang, Blair MacIntyre dl.acm.org/doi/10.1145/3613905.3651059 (CHI EA 2024)	May 2024
AvatarPilot: Decoupling One-to-One Motions from Their Semantics with Weighted Interpolations Cheng-Yao Wang, Eyal Ofek, Hyunju Kim, Payod Panda, Andrea Stevenson Won, Mar Gonzalez-Franco ieeexplore.ieee.org/document/10765138 (ISMAR Adjunct 2024)	Oct 2024
CollabXR: Bridging Realities in Collaborative Workspaces with Dynamic Plugin and Collaborative Tools Integration Cheng-Yao Wang, David Saffo, Bill Moriarty, Blair MacIntyre ieeexplore.ieee.org/document/10536279 (IEEE VRW 2024)	Mar 2024
Embodying Physics-Aware Avatars in Virtual Reality Yujie Tao, Cheng-Yao Wang, Andrew D. Wilson, Eyal Ofek, Mar Gonzalez-Franco dl.acm.org/doi/10.1145/3544548.3580979 (CHI 2023)	Apr 2023
VideoPoseVR: Authoring Virtual Reality Character Animations with Online Videos Cheng-Yao Wang, Qian Zhou, George Fitzmaurice, Fraser Anderson dl.acm.org/doi/10.1145/3567728 (PACM HCI (ISS 2022))	Nov 2022
CityLifeSim: A High-Fidelity Pedestrian and Vehicle Simulation with Complex Behaviors Cheng-Yao Wang, Oron Nir, Sai Vemprala, Ashish Kapoor, Eyal Ofek, Daniel McDuff, Mar Gonzalez-Franco ieeexplore.ieee.org/document/10070899 (IEEE ICIR 2022)	Dec 2022
Shared Realities: Avatar Identification and Privacy Concerns in Reconstructed Experiences Cheng-Yao Wang, Sandhya Sriram, Andrea Stevenson Won dl.acm.org/doi/10.1145/3476078 (PACM HCI (CSCW 2021))	Oct 2021
Hide and Seek: Choices of Virtual Backgrounds in Video Chats and Their Effects on Perception Angel Hsing-Chi Hwang, Cheng-Yao Wang, Yao-Yuan Yang, Andrea Stevenson Won dl.acm.org/doi/10.1145/3476044 (PACM HCI (CSCW 2021))	Oct 2021
Again, Together: Socially Reliving Virtual Reality Experiences When Separated Cheng-Yao Wang, Mose Sakashita, Upol Ehsan, Jingjin Li, Andrea Stevenson Won dl.acm.org/doi/10.1145/3313831.3376642 (CHI 2020)	Apr 2020
ReliveReality: Enabling Socially Reliving Experiences in Virtual Reality via a Single RGB Camera Cheng-Yao Wang, Shengguang Bai, Andrea Stevenson Won ieeexplore.ieee.org/document/9090534 (IEEE VRW 2020)	May 2020

Movebox: Democratizing Mocap for the Microsoft Rocketbox Avatar Library

Dec 2020

Mar Gonzalez-Franco, Zelia Egan, Matthew Peachey, Angus Antley, Tanmay Randhavane, Payod Panda, Yaying Zhang, Cheng-Yao Wang, Derek F. Reilly, Tabitha C. Peck

ieeexplore.ieee.org/document/9319096 (IEEE AIVR 2020)

Drone.io: A Gestural and Visual Interface for Human-Drone Interaction

Mar 2019

Jessica R. Cauchard, Alex Tamkin, Cheng-Yao Wang, Luke Vink, Michelle Park, Tommy Fang, James A. Landay

ieeexplore.ieee.org/document/8673011 (HRI 2019)

RoMA: Interactive Fabrication with Augmented Reality and a Robotic 3D Printer

Apr 2018

Huaishu Peng, Jimmy Briggs, Cheng-Yao Wang, Kevin Guo, Joseph Kider, Stefanie Mueller, Patrick Baudisch, François Guimbretière

dl.acm.org/doi/10.1145/3173574.3174153 (CHI 2018)

PalmGesture: Using Palms as Gesture Interfaces for Eyes-Free Input

Aug 2015

Cheng-Yao Wang, Min-Chieh Hsiu, Po-Tsung Chiu, Chiao-Hui Chang, Liwei Chan, Bing-Yu Chen, Mike Y. Chen

dl.acm.org/doi/10.1145/2785830.2785885 (MobileHCI 2015)

PalmType: Using Palms as Keyboards for Smart Glasses

Aug 2015

Cheng-Yao Wang, Wei-Chen Chu, Po-Tsung Chiu, Min-Chieh Hsiu, Yih-Harn Chiang, Mike Y. Chen

dl.acm.org/doi/10.1145/2785830.2785886 (MobileHCI 2015)

Evertutor: Automatically Creating Interactive Guided Tutorials on Smartphones by User Demonstration

Apr 2014

Cheng-Yao Wang, Wei-Chen Chu, Hou-Ren Chen, Chun-Yen Hsu, Mike Y. Chen

dl.acm.org/doi/10.1145/2556288.2557407 (CHI 2014)